CUC City University of Cairo	**Faculty of Engineering** **Basic Sciences Department** **Course Name: Mathematics 1** **Course Code: BAS111** **Examiner: Dr. Zaher Hassanein &** **Dr. Mahmoud El-Shaffey**	**Academic Year: 2025/2026** **Semester: Fall** **Level: Freshman** **Date: 10/11/2025** **Time: 60 min.** **Total Marks: 25 Marks**
	Midterm Exam	
ID:	**Name:**	**The total mark of student:**

Remark: - The Exam Consists of **TWO Questions** in **THREE Pages.**

Question (1): [12 Marks]

Find $\frac{dy}{dx}$ the first derivative of the following questions **(Choose 4 parts)**

i) $y = e^x + x^e + x^4 + 4^x + \sin^{-1} 4$ **(3 Marks)**

ii) $y = cosh^{-1}(x-1) + sec^{-1}(\ln x)$ **(3 Marks)**

iii) $y = \frac{(x^2-1)(x-1)^3}{\sqrt{x}}$ **(3 Marks)**

iv) $x^2 y + sinh y - tanh y = 7$ **(3 Marks)**

v) $y = log_3(x^2 - 5) + tan^{-1} x$ **(3 Marks)**

vi) $f(x) = (x^2+1)^{tan\,2}$ **(3 Marks)**

Question (2): [13 Marks]

(a) Calculate **THREE PARTS** of the following

i) Evaluate $\int \frac{xdx}{4-x^2}$ **(3 Marks)**

ii) Calculate the nth derivatives of the function $f(x) = (x^2 + 1)e^{3x}$ **(3 Marks)**

iii) $\lim_{x\to 0}\left(\frac{1}{sinx} - \frac{1}{\tan x}\right)$ **(3 Marks)**

iv) $\lim_{x\to 0}(1 - 3x)^{\frac{1}{4x}}$ **(3 Marks)**

(b) Show that the Taylor expansion of $f(x) = \cos x$ at $x = 0$ is given by $cos\, x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$ and use this expansion to approximate the value of $\cos\left(\frac{\pi}{4}\right)^2$. **(4 Marks)**

GOOD LUCK

Committee of Examiners:1-Dr. Zaher Hassanein

2- Dr. Mahmoud El-Shaffey